

LOSO Experiment B_accel_Istm

LSTM only · Accelerometers only (both hands), no Nothing

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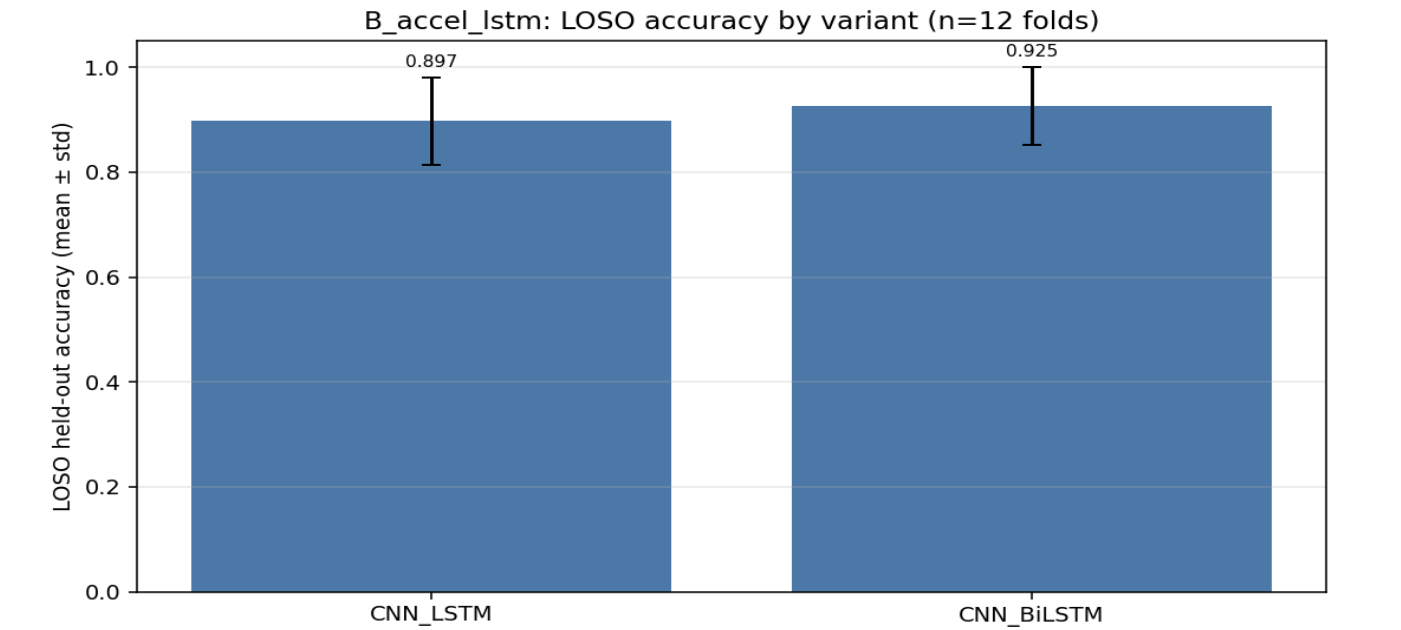
Experiment configuration

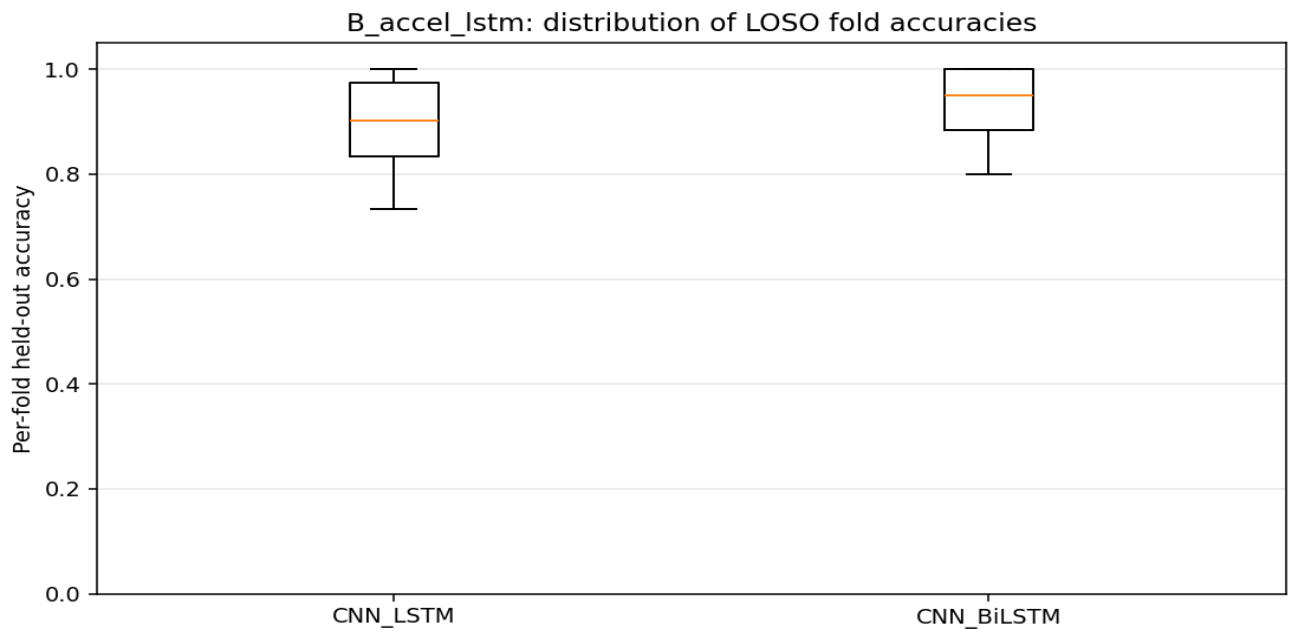
Class set	6 classes excluding Double_Nothing
Classes	Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor modalities	accel
Hands	left, right
Sensor channels	78
Sequence length	90 steps (resampled)
Filter	Butterworth 10.0Hz order 4
Normalisation	minmax
Augmentation	amplitude_scale + baseline_offset, 3 copies
Inner CV folds	3
Epochs · batch size	8 · 16
LOSO subjects	12: 11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry

LOSO summary across all subjects

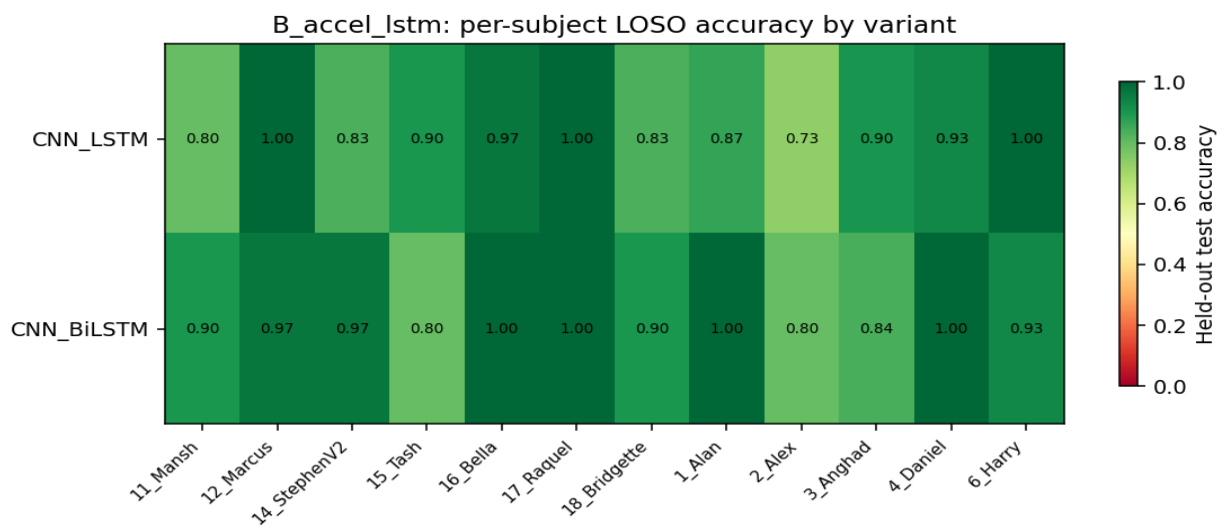
Variant	Params	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean (avg across folds)
CNN_LSTM	55,846	0.8975	0.0833	0.7333	1.0000	0.8886
CNN_BiLSTM	92,966	0.9254	0.0742	0.8000	1.0000	0.9448

Best variant by LOSO mean: **CNN_BiLSTM**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	CNN_LSTM	CNN_BiLSTM
11_Mansh	331	0.800	0.900
12_Marcus	331	1.000	0.967
14_StephenV2	331	0.833	0.967
15_Tash	331	0.900	0.800
16_Bella	331	0.967	1.000
17_Raquel	331	1.000	1.000
18_Bridgette	331	0.833	0.900
1_Alan	331	0.867	1.000
2_Alex	331	0.733	0.800
3_Anghad	330	0.903	0.839
4_Daniel	331	0.933	1.000
6_Harry	331	1.000	0.933

Variant: CNN_LSTM

Params: 55,846

LOSO mean accuracy: 0.8975 ± 0.0833 (min 0.7333, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	0/5	5/5	4/5	5/5	5/5	0.800
12_Marcus	5/5	5/5	5/5	5/5	5/5	5/5	1.000
14_StephenV2	5/5	5/5	0/5	5/5	5/5	5/5	0.833
15_Tash	3/5	5/5	5/5	5/5	5/5	4/5	0.900
16_Bella	5/5	5/5	5/5	4/5	5/5	5/5	0.967
17_Raquel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
18_Bridgette	0/5	5/5	5/5	5/5	5/5	5/5	0.833
1_Alan	5/5	5/5	5/5	1/5	5/5	5/5	0.867
2_Alex	5/5	4/5	1/5	2/5	5/5	5/5	0.733
3_Anghad	5/5	5/5	5/5	2/5	6/6	5/5	0.903
4_Daniel	5/5	3/5	5/5	5/5	5/5	5/5	0.933
6_Harry	5/5	5/5	5/5	5/5	5/5	5/5	1.000
MEAN	0.883	0.867	0.850	0.800	1.000	0.983	0.897

Variant: CNN_BiLSTM

Params: 92,966

LOSO mean accuracy: 0.9254 ± 0.0742 (min 0.8000, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Lower	Double_Pistol_Recoil	Double_Raise	Double_Wiggle	Double_cmere	Drum_Roll	Overall
11_Mansh	5/5	2/5	5/5	5/5	5/5	5/5	0.900
12_Marcus	5/5	5/5	5/5	5/5	5/5	4/5	0.967
14_StephenV2	5/5	5/5	5/5	4/5	5/5	5/5	0.967
15_Tash	3/5	5/5	5/5	5/5	5/5	1/5	0.800
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
18_Bridgette	2/5	5/5	5/5	5/5	5/5	5/5	0.900
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	4/5	1/5	5/5	5/5	4/5	0.800
3_Anghad	5/5	5/5	5/5	5/5	1/6	5/5	0.839
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	5/5	5/5	5/5	3/5	5/5	0.933
MEAN	0.917	0.933	0.933	0.983	0.897	0.900	0.925